

Uplink Resource Allocation for eMBB-URLLC Coexistence in Multi-UAV Networks

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UAV-Assisted Multiplexing of eMBB and URLLC

- We consider a **multi-UAV assisted uplink communication network** supporting the coexistence of **eMBB and URLLC services**.
- In the considered system, a set of ground users transmit uplink data to aerial access points (UAVs), which are ABS. Each UAV serves a group of users through uplink wireless access links.
- To improve spectral efficiency, **NOMA** is employed so that an eMBB user and a URLLC user may share the same resource block.
- At the UAV receiver, **successive interference cancellation (SIC)** is used to separate the superposed signals.

- We denote the UAV set by

$$N = \{1, 2, \dots, n\}$$

- The eMBB user set

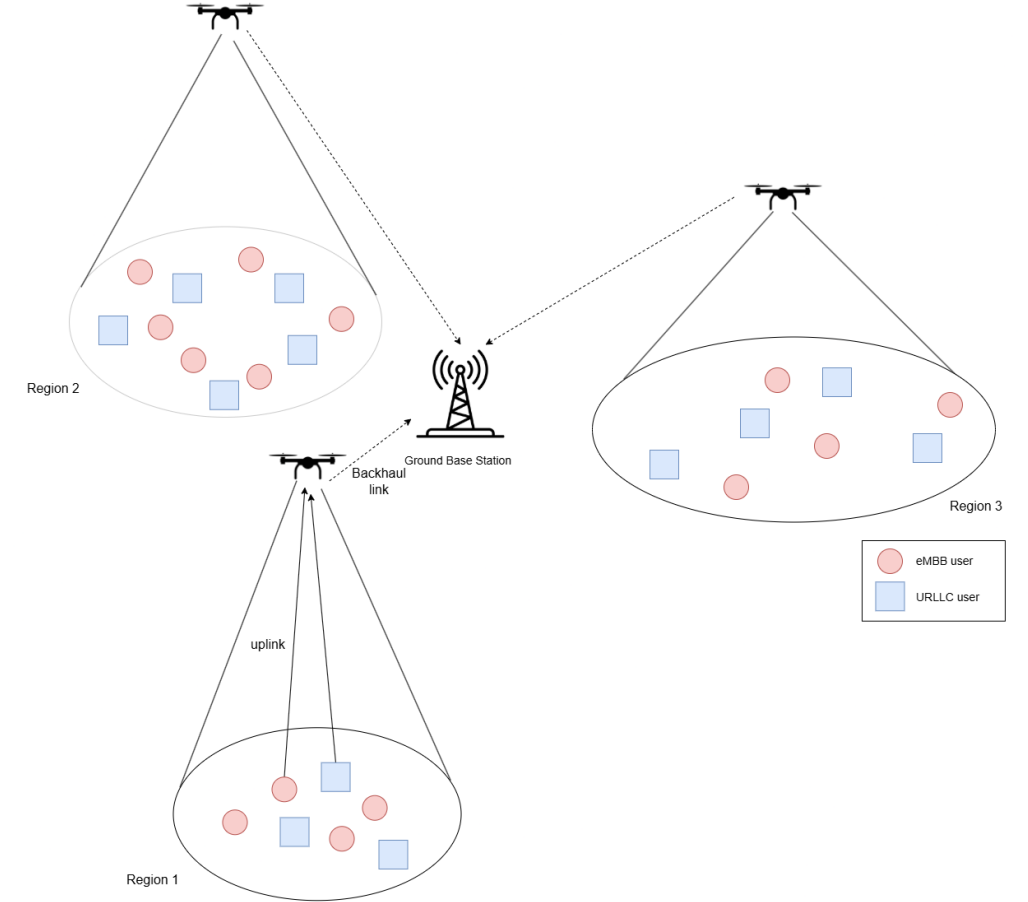
$$E = \{1, 2, \dots, e\}$$

- The URLLC user set

$$U = \{1, 2, \dots, u\}$$

- Let $\phi_{i,j}^t \in \{0, 1\}$ denote whether user i is associated with UAV j at slot t . Each user is associated with one UAV.

The time axis is divided into slots indexed by $t \in T$, and each slot is further partitioned into S minislots indexed by $s \in \mathcal{S} \triangleq \{1, 2, \dots, S\}$. The system bandwidth is divided into K RBs indexed by $k \in \mathcal{K} \triangleq \{1, 2, \dots, K\}$.



Propagation Model

- For the uplink transmission from a ground user $i \in E \cup U$ to UAV $j \in N$, the air-to-ground (A2G) channel is modeled as a probabilistic LoS/NLoS channel.

- The distance between ground user i and UAV j is given by

$$d_{i,j}(t) = \sqrt{(d_{i,j}^h(t))^2 + H_j^2}$$

- The elevation angle is defined as

$$\theta_{i,j}(t) = \frac{180}{\pi} \arctan\left(\frac{H_j}{d_{i,j}^h(t)}\right)$$

- Following the commonly used probabilistic A2G model, the LoS

probability is given by

$$P_{i,j}^{LoS}(t) = \frac{1}{1 + a \exp[-b(\theta_{i,j}(t) - a)]}.$$

- Hence, the non-line-of-sight (NLoS) probability is

$$P_{i,j}^{NLoS}(t) = 1 - P_{i,j}^{LoS}(t).$$

- The path loss under LoS and NLoS conditions is written as

$$PL_{i,j}^{LoS}(t) = FSPL_{i,j}(t) + \eta_{LoS}.$$

$$PL_{i,j}^{NLoS}(t) = FSPL_{i,j}(t) + \eta_{NLoS}.$$

- Where

$$FSPL_{i,j}(t) = 20 \log_{10}(d_{i,j}(t)) + 20 \log_{10}(f_c) + 20 \log_{10}\left(\frac{4\pi}{c}\right).$$

The average large-scale path loss is

$$\overline{PL}_{i,j}(t) = P_{i,j}^{LoS}(t) PL_{i,j}^{LoS}(t) + (1 - P_{i,j}^{LoS}(t)) PL_{i,j}^{NLoS}(t).$$

The corresponding large-scale channel gain is

$$\beta_{i,j}(t) = 10^{-\overline{PL}_{i,j}(t)/10}.$$

Finally, the channel coefficient is modeled as

$$h_{i,j}(t) = \sqrt{\beta_{i,j}(t)} g_{i,j}(t).$$

where $g_{i,j}$ denotes the small-scale fading coefficient.

For analytical tractability, we assume block fading within one time slot and let

$$g_{i,j}(t) \sim \mathcal{CN}(0,1).$$

Symbol	Description
i	Ground user index
j	UAV index
$d_{i,j}^h(t)$	Horizontal distance between ground user i and UAV j at time slot t
H_j	Altitude of UAV j
$d_{i,j}(t)$	Three-dimensional distance between ground user i and UAV j at time slot t
$\theta_{i,j}(t)$	Elevation angle from ground user i to UAV j
$P_{i,j}^{LoS}(t)$	Probability that the A2G link between user i and UAV j is line-of-sight
$P_{i,j}^{NLoS}(t)$	Probability that the A2G link between user i and UAV j is non-line-of-sight
$\beta_{i,j}(t)$	Large-scale channel gain between user i and UAV j
$h_{i,j}(t)$	Complex channel coefficient of the A2G link
$g_{i,j}(t)$	Small-scale fading coefficient

Signal Model and Coexistence Modes

- **Receive signal**

$$\zeta_j = \sum_{q \in E} \sqrt{p_q^t} h_{q,j}^t \mu_q + \sum_{z \in U} \sqrt{p_z^{s,t}} h_{z,j}^t \mu_z + n_j$$

- The received signal is the superposition of:
 - uplink signals from eMBB users
 - uplink signals from URLLC users
 - $n_j(t) \sim \mathcal{CN}(0, \sigma_j^2)$ is the additive receiver noise.

- **Superposition and Puncturing Modes**

- Superposition Mode

- Under superposition mode, the URLLC user is multiplexed with an ongoing eMBB user on the same RB. This uplink non-orthogonal transmission corresponds to the received signal model in (above)
 - We define the superposition indicator by

$$\rho_{q,z,k,j}^{s,t} \in \{0,1\}$$

- Puncturing Mode

- Under puncturing mode, the ongoing eMBB transmission is suspended in minislot s , and the RB is exclusively used by the URLLC user. We define the puncturing indicator by

$$\varrho_{q,z,k,j}^{s,t} \in \{0,1\}$$

- Since one RB in one minislot can only be operated in one coexistence mode

$$\rho_{q,z,k,j}^{s,t} + \varrho_{q,z,k,j}^{s,t} = 1$$

SIC Decoding Order

- At the UAV receiver, SIC is adopted to decode the superposed uplink signals. Since URLLC traffic is latency-critical, the URLLC packet is decoded before the eMBB packet.
- Hence, the decoding order is:
 1. Decode the URLLC signal first;
 2. If the URLLC signal is successfully decoded, subtract it from the received signal;
 3. Decode the eMBB signal from the residual signal.

Symbol	Description
j	UAV index
$\zeta_j(t)$	received signal at UAV j at time slot t
E	set of eMBB users
U	set of URLLC users
p_q^t	transmit power of eMBB user q
$p_z^{s,t}$	transmit power of URLLC user z
$h_{q,j}^t$	channel coefficient from eMBB user q to UAV j
$h_{z,j}^t$	channel coefficient from URLLC user z to UAV j
μ_q	transmitted symbol of eMBB user q
μ_z	transmitted symbol of URLLC user z
n_j	additive white Gaussian noise at UAV j

SINR Model

- **URLLC Decoding**

- **URLLC SINR model with mode-switching**

$$\delta_{z,j,k}^{s,t} = \frac{p_z^{s,t} |h_{z,j}^t|^2}{\rho_{q,z,k,j}^{s,t} p_q^t |h_{q,j}^t|^2 + I_{j,k}^{s,t} + \sigma_j^2}$$

- **eMBB Decoding**

- **eMBB SINR model considering SIC residual and puncturing**

$$\delta_{q,j,k}^{s,t} = (1 - \varrho_{q,z,k,j}^{s,t}) \frac{p_q^t |h_{q,j}^t|^2}{\eta_{sic} p_z^{s,t} |h_{z,j}^t|^2 + I_{j,k}^{s,t} + \sigma_j^2}$$

Aggregate Inter-Cell Interference

$$I_{j,k}^{s,t}$$

$$= \sum_{j' \in N, j' \neq j} \left(\sum_{q \in E} \phi_{q,j'}^t (1 - \varrho_{q,z,k,j'}^{s,t}) p_q^t |h_{q,j}^t|^2 + \sum_{z \in U} \phi_{z,j'}^t p_z^{s,t} |h_{z,j}^t|^2 \right)$$

Symbol	Description
j	UAV index
z	URLLC user index
q	eMBB users index
$p_z^{s,t}$	transmit power of URLLC user z
$h_{z,j}^t$	channel coefficient from URLLC user z to UAV j
p_q^t	transmit power of eMBB user q
$h_{q,j}^t$	channel coefficient from eMBB user q to UAV j
I_j	Aggregate Inter-Cell Interference
σ_j^2	noise power at UAV j
η_{sic}	Residual interference level (factor) from the URLLC user after cancellation

Rate Model

- **URLLC rate model**

- By converting the finite-blocklength decoding requirement into a minimum SINR condition, the target SINR threshold for URLLC user z is approximated as

$$\Gamma_z^{th} \approx 2^{\frac{L_z}{\tau B_k} + \sqrt{\frac{V_0}{m_{z,k}^{s,t}} Q^{-1}(\epsilon_z)}} - 1$$

- where L_z is the URLLC payload size, τ is the minislot duration, B_k is the RB bandwidth, $m_{z,k}^{s,t}$ is the blocklength, ϵ_z is the target decoding error probability, and V_0 is the approximated channel dispersion term.

- **eMBB rate model**

$$r_{q,j,k}^{s,t} = B_k \log_2(1 + \delta_{q,j,k}^{s,t})$$

Let $\alpha_{q,j,k}^E$ be the RB allocation indicator at time slot t , defined as:

$$\alpha_{q,j,k}^E = \begin{cases} 1, & \text{if RB } k \text{ at UAV } j \text{ is allocated to eMBB user } q \\ 0, & \text{otherwise} \end{cases}$$

$$R_{q,j}^{s,t} = \sum_{k \in K} \alpha_{q,j,k}^E r_{q,j,k}^{s,t}$$

Symbol	Description
q	Index of an eMBB user
z	Index of a URLLC user
j	Index of UAV (serving access point)
k	Index of resource block (RB)
s	Index of minislot
t	Time slot index
L_z	Payload size of the URLLC packet of user z
τ	Mini-slot duration
B_k	Bandwidth of resource block (k)
$\delta_{z,j,k}^{s,t}$	SINR of the URLLC user
$\delta_{z,j,k}^{s,t,sp}$	SINR of URLLC user under superposition transmission
$\delta_{z,j,k}^{U,pun}$	SINR of URLLC user under puncturing
$\delta_{q,j,k}^{s,t,SIC}$	SINR of eMBB user after successful SIC
$\delta_{q,j,k}^{s,t,res}$	SINR of eMBB user with imperfect SIC (residual interference)
$\rho_{q,z,k,j}^{s,t}$	Superposition indicator (1 if URLLC and eMBB share the RB, otherwise 0)
$\varrho_{q,z,k,j}^{s,t}$	Puncturing indicator (1 if URLLC punctures the RB of eMBB, otherwise 0)
V_0	Channel dispersion of the URLLC transmission
$m_{z,k}^{s,t}$	Blocklength of the URLLC packet
ϵ_z	Target decoding error probability of URLLC
$Q^{-1}()$	Inverse Gaussian Q-function

Problem formulation

$$\max_{\alpha, \rho, \varrho, p} \sum_{t \in T} \sum_{s \in S} \sum_{j \in N} \sum_{q \in E} R_{q,j}^{s,t}$$

$$\text{s.t.} \quad x_j^{\min} \leq x_j \leq x_j^{\max}, y_j^{\min} \leq y_j \leq y_j^{\max}, H_j^{\min} \leq H_j \leq H_j^{\max} \quad \forall j, t$$

$$\sum_{j \in N} \phi_{i,j}^t = 1 \quad \forall i, t$$

$$\delta_{z,j,k}^{s,t} > \Gamma_z^{th} \quad \forall z$$

$$R_{q,j}^{s,t} \geq R_{q,j,\min}^{s,t} \quad \forall q, j$$

$$0 \leq p_q \leq p_q^{\max} \quad \forall q$$

$$0 \leq p_z \leq p_z^{\max} \quad \forall z$$

$$\sum_{q \in E} \alpha_{q,j,k}^E = 1 \quad \forall k, j, t$$

$$\rho_{q,z,k,j}^{s,t} \leq \alpha_{q,k,j}^t, \quad \varrho_{q,z,k,j}^{s,t} \leq \alpha_{q,k,j}^t \quad \forall q, z, k, j, s, t$$

$$\rho_{q,z,k,j}^{s,t} + \varrho_{q,z,k,j}^{s,t} = 1 \quad \forall q, z, k, j, s, t$$

$$\sum_{q \in E} \sum_{z \in U} (\rho_{q,z,k,j}^{s,t} + \varrho_{q,z,k,j}^{s,t}) = 1 \quad \forall k, j, s, t$$

$$\sum_{k \in \mathcal{K}} \sum_{q \in E} \sum_{z \in U} (\rho_{q,z,k,j}^{s,t} + \varrho_{q,z,k,j}^{s,t}) = |\mathcal{K}| \quad \forall j, s, t$$

$$\alpha_{q,j,k}^E, \rho_{q,u,k,j}^{s,t}, \varrho_{q,u,k,j}^{s,t} \in \{0,1\}$$

Symbol	Description
q	Index of an eMBB user
z	Index of a URLLC user
j	Index of UAV (serving access point)
k	Index of resource block (RB)
s	Index of minislot
t	Time slot index
E	Set of eMBB users
U	Set of URLLC users
N	Set of UAV
$\mathcal{K}$	Set of RBs
S	Set of minislots
$R_{q,j}^{s,t}$	achievable rate of eMBB user q served by UAV j
$R_{z,j}^t$	achievable rate of URLLC user z served by UAV j
L_z	Payload size of the URLLC packet of user z
P_e	Decoding error probability of URLLC user z
ϵ_z	tolerable decoding error probability of URLLC user z
$R_{q,j,\min}^{s,t}$	Minimum required rate of eMBB user q
p_q	Transmit power of eMBB user q in slot t
p_z	Transmit power of URLLC user z in slot t
$\alpha_{q,j,k}^E$	RB allocation indicator for eMBB user q on RB k of UAV j in slot t
$\rho_{q,z,k,j}^{s,t}$	Superposition indicator (1 if URLLC and eMBB share the RB, otherwise 0)
$\varrho_{q,z,k,j}^{s,t}$	Puncturing indicator (1 if URLLC punctures the RB of eMBB, otherwise 0)
τ	Mini-slot duration